A SECURE BLOCKCHAIN-BASED CERTIFICATE VERIFICATION FRAMEWORK USING FACIAL BIOMETRICS AND HYBRID CRYPTOGRAPHY

A PROJECT REPORT

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TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	**vii**
	PROBLEM STATEMENT	**viii**
	LIST OF FIGURES	**ix**
	LIST OF ABBREVIATION	**X**
1	**INTRODUCTION**	**1**
	1.1 BACKGROUND OF THE PROJECT	1
	1.2 MOTIVATION	2
	1.3 PROPOSED SOLUTION	3
	1.4 OBJECTIVES	4
	1.5 SCOPE	5
2	**LITERATURE REVIEW**	**6**
3	**SYSTEM ANALYSIS**	**16**
	3.1 EXISTING SYSTEM	16
	3.1.1 Disadvantages of Existing System	17
	3.2 PROPOSED SYSTEM	17
	3.2.1 Advantages of Proposed System	18
4	**SYSTEM SPECIFICATION**	**19**
	4.1 HARDWARE REQUIREMENTS	19
	4.2 SOFTWARE REQUIREMENTS	19
5	**SOFTWARE DESCRIPTION**	**20**
	5.1 FRONTEND DEVELOPMENT	20
	5.2 BACKEND DEVELOPMENT	22

	5.3 API TOOLS AND TESTING	25
	5.4 PYPHARM	26
	5.5 OPERATING SYSTEM	30
6	**SYSTEM DESIGN**	**31**
	6.1 SYSTEM ARCHITECTURE	31
	6.2 UML DIAGRAM	32
	6.2.1 Use Case Diagram	33
	6.2.2 Class Diagram	34
	6.2.3 Sequence Diagram	35
	6.2.4 Deployment Diagram	36
	6.2.5 Data Flow Diagram	37
7	**PROJECT DESCRIPTION**	**42**
	7.1 DESCRIPTION	42
	7.2 MODULES DESCRIPTION	42
	7.2.1 FRAME CREATION	42
	7.2.2 INSTITUTION PROCESS	43
	7.2.3 FILE ENCRYPTION	43
	7.2.4 BLOCK CHAIN STORAGE	44
	7.2.5 USER AUTHENTICATION	44
	7.2.6 DECRYPTION AND DOWNLOAD FILE	44
	7.2.7 VERIFIER PROCESS	45
	7.3 ALGORITHM	45
	7.3.1 GRASSMANN ALGORITHM	45
	7.3.2 ADVANCED ENCRYPTION STANDARD	46
	7.3.3 ELLIPTIC CURVE CROYPTOGRAPHY	46
	7.3.4 BLOCKCHAIN TRANSACTION AND STORAGE ALGORITHM	46

8	**SYSTEM DESIGN**	**47**
	8.1 TESTING OBJECTIVES	47
	8.2 TYPES OF TESTING	48
	8.2.1 UNIT TESTING	48
	8.2.2 INTERGRATION TESTING	48
	8.2.3 PERFORMANCE TESTING	48
	8.2.4 SECURITY TESTING	48
	8.2.5 USABILITY TESTING	48
9	**CONCLUSION AND FUTURE ENHANCEMENT**	**49**
	9.1 CONCLUSION	49
	9.2 FUTURE ENHANCEMENT	50
10	**APPENDIX**	**51**
	10.1 SOURCE CODE	51
	10.2 SCREENSHOTS	65
11	**REFERENCES**	**69**

ABSTRACT

The rapid growth of digital technologies has increased the need for secure and reliable certificate verification systems. Traditional methods of certificate validation are often manual, time-consuming, and prone to forgery and data tampering. To address these issues, this project proposes a secure blockchain-based certificate verification framework using facial biometrics and hybrid cryptography**.**The system utilizes blockchain technology to store certificate data in a decentralized and immutable manner, ensuring that once the data is recorded, it cannot be altered or manipulated. To enhance security, hybrid cryptography techniques, combining Advanced Encryption Standard (AES) and Elliptic Curve Cryptography (ECC), are used for encrypting and decrypting certificate data. This ensures both high security and efficient performance.

Additionally, the integration of facial biometric authentication provides a reliable method for verifying the identity of users, preventing unauthorized access. The system also incorporates One-Time Password (OTP) verification as an extra layer of security.The proposed framework enables educational institutions to upload and manage certificates securely, while users and verifiers can easily access and validate them in real-time.

PROBLEM STATEMENT

In the current system, certificate verification is primarily carried out through manual processes or centralized digital platforms, which are inefficient and vulnerable to security threats. One of the major issues is the increasing number of fake and forged certificates, which leads to serious problems in recruitment and academic validation. Organizations often struggle to verify the authenticity of certificates due to the lack of a unified and secure verification mechanism.Moreover, existing systems lack strong data protection measures, making them susceptible to data breaches, unauthorized access, and tampering. Centralized storage systems are prone to single points of failure, where any compromise can affect the entire database. The absence of advanced authentication mechanisms further increases the risk of identity fraud.

Another key challenge is the time-consuming verification process, which involves multiple intermediaries and manual validation steps. This leads to delays and inefficiencies, especially when dealing with large volumes of certificates.Therefore, there is a need for a secure, decentralized, and automated certificate verification system that ensures data integrity, prevents forgery, and provides fast and reliable authentication.

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
6.1	System Architecture	28
6.2.1	Use Case Diagram	30
6.2.2	Class Diagram	31
6.2.3	Sequence Diagram	32
6.2.4	Deployment Diagram	33
6.2.5	Data Flow Diagram level 0	34
6.2.6	Data Flow Diagram level 1	35
6.2.7	Data Flow Diagram level 2	36
6.2.8	Data Flow Diagram level 3	55
6.2.9	Data Flow Diagram level 4	55
10.1	Home Page	73
10.2	New User Register	73
10.3	Verifier Page	74
10.4	User Login Page	74
10.5	OTP Verification Page	75
10.6	Personal Information Page	75
10.7	Document Information Page	76
10.8	College Login Page	76

LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSIONS
AES	Advanced Encryption Standard
ECC	Elliptic Curve Cryptography
AI	Artificial Intelligence
OTP	One Time Password
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
DOM	Document Object Model
SQL	Structured Query Language
DB	Database